

AGENTIC AI: FROM BUSINESS NEEDS TO WORKING AGENTS

Professor Peyman Teymoori

**BUSINESS
NEEDS**

WORKING AGENTS



Designing Agentic AI: *From Need to Spec*

Professor Peyman Teymoori

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Kickoff & Foundations





From Business Pain to Agent Plan

01

Lecture Overview

This lecture transforms business problems into clear, testable agent designs.

02

Learning Outcomes

define what makes software agentic, translate business problems into measurable KPIs, design system instructions, and create specific, testable success criteria for agent behavior.

03

Mindset Shift

Adopt a mindset where agents augment human capabilities rather than replace them.

Learn to start with constrained designs and expand carefully, ensuring safety and effectiveness at every step.

What Makes Software Agentic?

Traditional vs. Agentic Software

Traditional software follows a fixed logic:

input → logic → output.

Agentic software pursues goals through a dynamic loop:

perceive → plan → act → evaluate → adjust.

Key Characteristics

An agent is goal-driven,

perceives its environment,

makes autonomous decisions,

uses tools to affect its environment,

and learns from feedback.



Core Agent Loop Explained



Perceive

The agent gathers inputs, memory, and context to understand the current state. This step is crucial for making informed decisions based on real-time data and historical information.

Act

The agent invokes tools and updates its state based on the planned actions. This step is where the agent interacts with the environment to make progress toward its goal.

Plan

The agent breaks down the goal into actionable steps and selects the appropriate tools. Planning ensures that the agent has a clear roadmap to achieve its objective.

Evaluate

The agent assesses whether the goal has been achieved and triggers replanning if necessary. Evaluation ensures continuous improvement and adaptation to changing conditions.



Single vs Multi-Agent Choice

Choosing the Right Approach

- Use a single agent for focused tasks with limited tools and sequential workflows.
- Consider multi-agent systems for complex tasks requiring distinct expertise, parallel work, and coordination.
- Start with a single agent and split only when complexity justifies it.



Real-World Inspiration



“ ...

Healthcare Intake Assistant

...

01

Business Problem

A clinic receives 200+ intake forms daily, leading to manual review bottlenecks and frequent insurance verification errors, delaying appointments and increasing operational costs.

02

Agent Solution

The agent extracts data, verifies insurance, checks network status, and auto-schedules 70% of cases. It escalates complex cases to humans, ensuring only verified patients are booked.

03

Outcome

The agent reduced intake time by 65% and improved accuracy to 94%. By handling the 'happy path' cases, it freed up human resources to focus on more complex issues.

Investment Advisory Co-Pilot



01

Business Problem

Junior advisors spend over five hours preparing portfolio reviews, leading to repetitive analysis and inconsistent narrative quality, affecting client satisfaction.

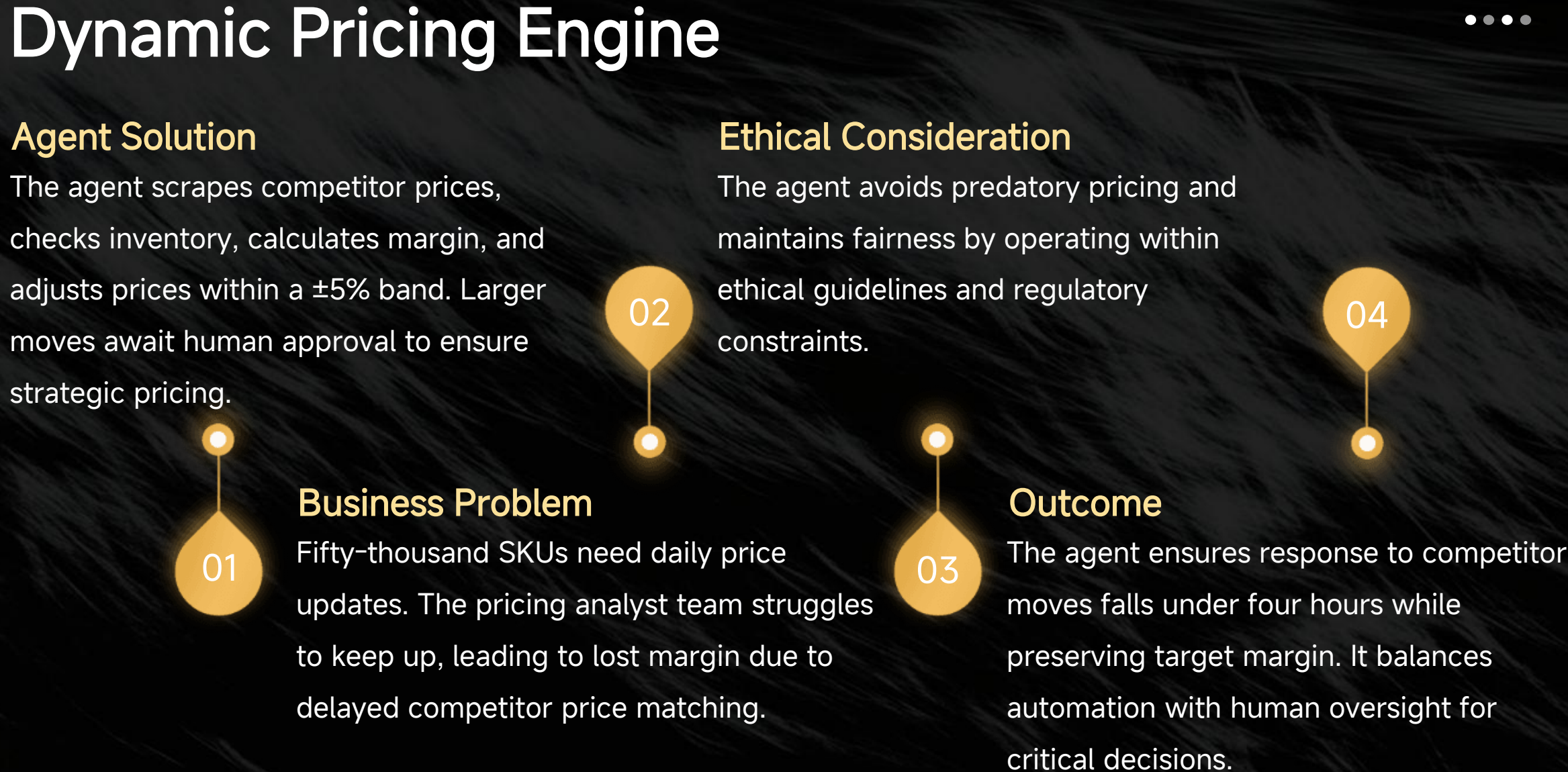
02

Agent Solution

The agent fetches holdings, calculates metrics, benchmarks performance, and drafts narrative sections. Advisors refine and own the final advice, ensuring compliance and personalized touch.



Dynamic Pricing Engine



Agent Solution

The agent scrapes competitor prices, checks inventory, calculates margin, and adjusts prices within a $\pm 5\%$ band. Larger moves await human approval to ensure strategic pricing.

Ethical Consideration

The agent avoids predatory pricing and maintains fairness by operating within ethical guidelines and regulatory constraints.

Business Problem

Fifty-thousand SKUs need daily price updates. The pricing analyst team struggles to keep up, leading to lost margin due to delayed competitor price matching.

Outcome

The agent ensures response to competitor moves falls under four hours while preserving target margin. It balances automation with human oversight for critical decisions.

Support Triage & Routing Bot



Business Problem and Solution

One-thousand daily tickets were mis-routed 40% of the time.

The agent classifies intent, scores sentiment, searches the knowledge base, and suggests replies.

Angry customers are prioritized for senior reps.

Routing accuracy jumps to 92% and first response shrinks to 1.5 hours.





Build Your Canvas





Agent Canvas at a Glance



Purpose

The Agent Canvas compresses the business case into one page, ensuring alignment on value and boundaries before technical work begins. It keeps the team aligned during implementation.

Components

It includes problem definition, target user, KPIs, inputs, outputs, data sources, constraints, scope, risks, and human-in-the-loop gates.

Set Measurable KPIs



Importance

01

KPIs drive prioritization and give evaluators clear pass-fail targets. They ensure the agent meets business goals and provides tangible value.

Examples

02

Accuracy >85%, response time <2 hours, cost <\$0.05 per query. Each KPI needs an owner and a measurement method to track progress.

Avoid Vagueness

03

Replace vague goals with specific numbers. 'Better' is not a KPI; '50% reduction in response time' is. Specific KPIs enable precise evaluation and continuous improvement.





Write the Spec



System Instruction Anatomy

...

01

Role Definition

Clearly define the agent's role and purpose. Use imperative language to specify its responsibilities and capabilities.

02

Constraints

Specify what the agent must and must not do. Constraints ensure safety, compliance, and ethical behavior.

03

Process

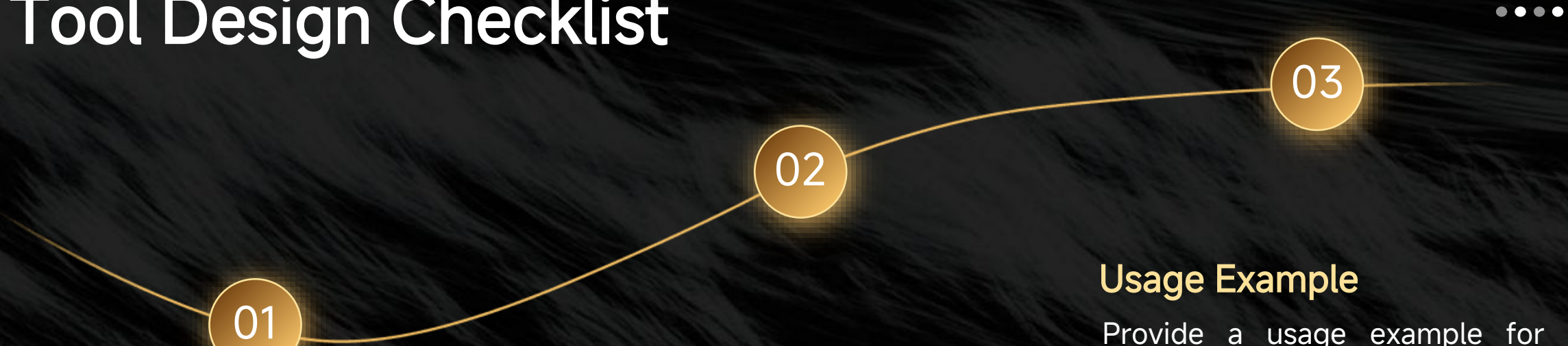
Outline the step-by-step process the agent follows. Clear steps reduce ambiguity and ensure consistent behavior.

04

Output Format

Define the format of the agent's output. This ensures consistency and makes it easier for users to understand and interact with the agent.

Tool Design Checklist



01

Tool Naming

Each tool needs a clear, descriptive name that reflects its purpose. This makes it easier for developers and users to understand its function.

02

Input and Output

Define the input schema and output schema for each tool. Specify what each field means and how it should be used.

03

Usage Example

Provide a usage example for each tool. This helps illustrate how the tool should be called and what outputs to expect.

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Memory Types & Policies

Memory Types

Define what type of memory the agent will use: working memory for current tasks, session memory for user preferences, and long-term memory for learned patterns.

Retention Policy

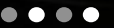
Specify when to store information and what to purge. Ensure compliance with privacy rules and data-retention policies to protect user data.



Craft Prompts



Prompt Evolution Workflow



01

Start Naive

Begin with a simple role statement. Then iteratively add capabilities, constraints, process, and examples. Each iteration improves clarity and safety.

Adversarial Testing

02

Run mental adversarial tests after each version to find gaps. This ensures the prompt reliably produces the desired behavior across diverse inputs.

03

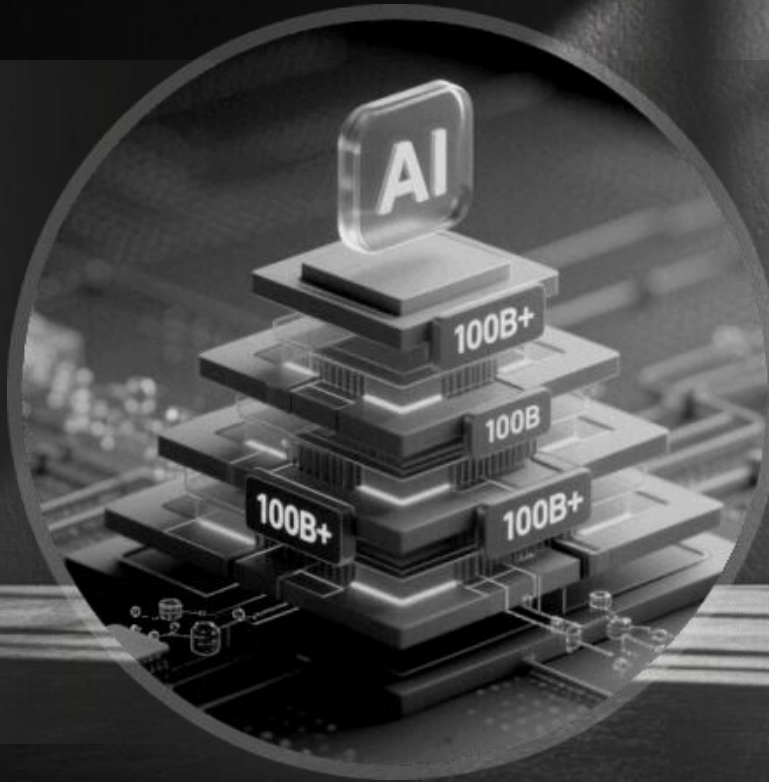
Iterate

Continue iterating until the prompt produces the desired structure, tone, and safety behavior without becoming unnecessarily long.

Guardrails in Language

Explicit Directives

Use explicit directives like 'MUST' and 'MUST NOT' to enforce safety and compliance. Specify output format with examples.



Embedded Safety

Embed guardrails directly in the prompt. This ensures they travel with the agent and are enforced wherever it is deployed.

Design Tests



Five Must-Pass Tests



01

Test Categories

Every agent needs at least one happy path, one edge case, one guardrail refusal, one error handling, and one safety-critical accuracy test. These tests define success.

02

Test Structure

Each test states input, expected behavior, pass criteria, and business reason. These tests become the definition of done and protect against regressions during iteration.



Happy Path & Edge Cases



Happy Path

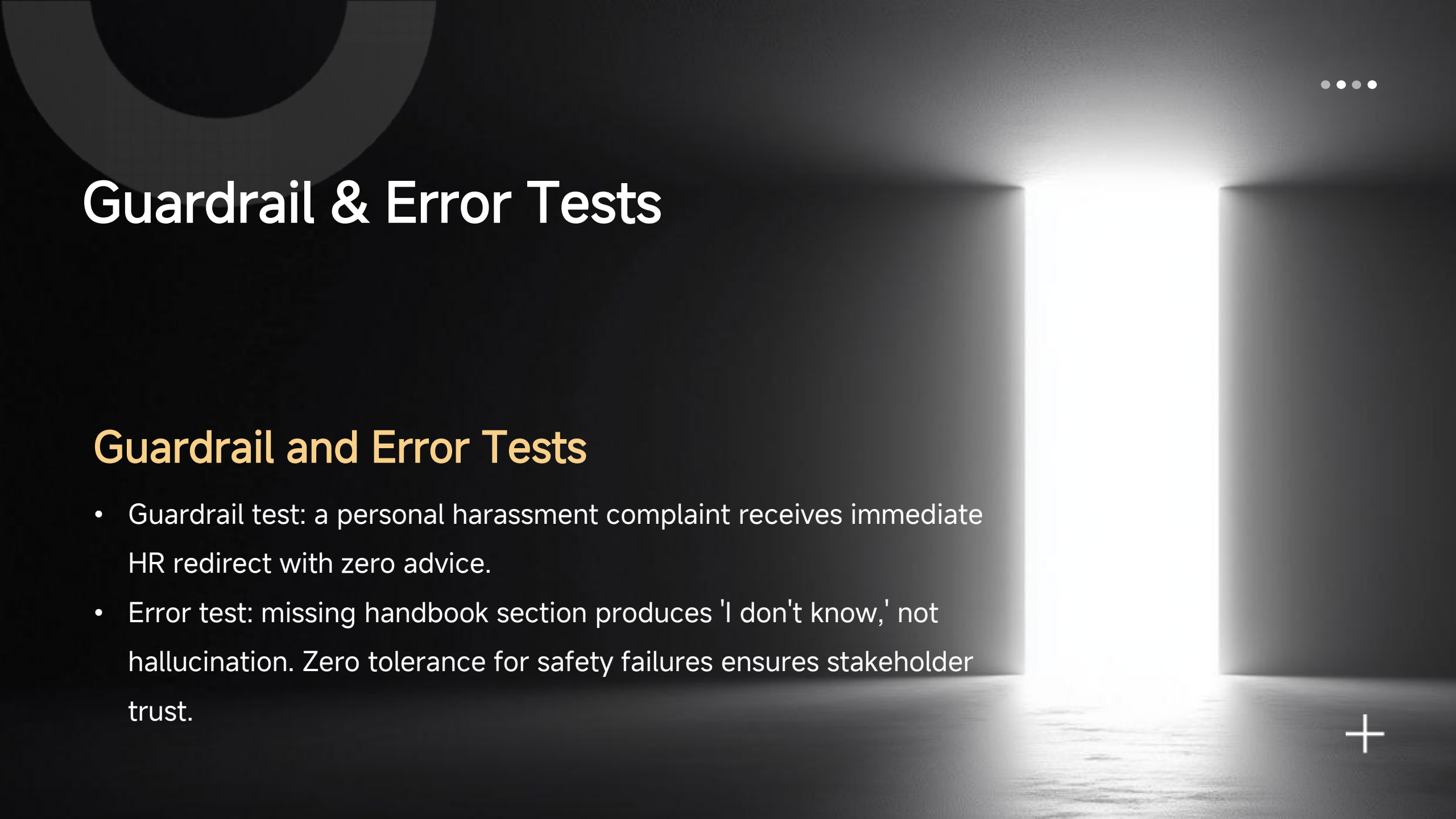
The happy path confirms core value. For example, a standard PTO question returns the right number with citation. This ensures the agent meets basic requirements.

Edge Cases

Edge cases probe boundaries. For example, an overly broad benefits query triggers clarification, not overload. This ensures the agent handles unusual inputs gracefully.

Importance

Both happy path and edge case tests must pass at high confidence before the agent meets real users. This ensures reliability and user satisfaction.



Guardrail & Error Tests

Guardrail and Error Tests

- Guardrail test: a personal harassment complaint receives immediate HR redirect with zero advice.
- Error test: missing handbook section produces 'I don't know,' not hallucination. Zero tolerance for safety failures ensures stakeholder trust.

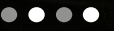




Review & Iterate



Peer Review Carousel



01

Review Process

Teams rotate reviewing each other's Canvas, Spec, and Tests. Highlight strengths, then offer one actionable improvement.

02

Feedback

External eyes catch blind spots in scope, risk, or feasibility and spread best practices across the cohort before implementation begins.

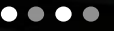
03

Importance

Peer review ensures diverse perspectives are considered, leading to more robust and reliable designs.



Workshop Activity



Brainstorm (5 min.)

Identify a business problem in your organization that could benefit from an agent

Design (10 min.)

Complete the Agent Canvas and draft your Agent Spec Sheet

Test (5 min.)

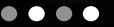
Create 5 test cases covering happy path, edge cases, and guardrails



Next Steps



Lecture 2 Preview



Next Session

Next session converts your spec into working code. We will choose an architecture pattern, implement your first tool, add memory, and run the five must-pass tests.



Preparation

Bring validated assumptions and updated documents to hit the ground running. Ensure you understand your rotating role and have access to shared documents.



Key Takeaways Today



Agent Characteristics

Agents pursue goals through perceive-plan-act loops, not fixed rules. This dynamic approach enables them to handle complex, variable tasks effectively.

Canvas and Spec

Business Canvas drives technical spec; guardrails and tests are core features, not add-ons. Start with a clear Canvas to ensure alignment and success.

Iteration Mindset

Start constrained, expand carefully. Peer review and early validation save time. Measurable KPIs define success and guide iteration.

Safety First

Guardrails and safety checks are essential. Ensure zero tolerance for safety failures to maintain stakeholder trust and compliance.

Further Resources

Access Resources

Access slide templates, example agents, and test banks in the course drive. Recommended reading includes 'Designing Agentic Systems' safety paper and OpenAI function-calling docs.

Online

<https://github.com/peyman-t/AgenticAI>



Thank You & Build Safe

Closing Remarks

You now hold a complete, reviewed blueprint for an agent that solves a real business pain. Guard your KPIs, honor your guardrails, and keep humans in the loop when uncertainty looms.

